

SCIENCE IS THE PATH TO KNOWLEDGE, AND FINDING OUT ABOUT HOW OUR UNIVERSE WORKS. YOU CAN'T BE A GEEK AND NOT KNOW YOUR SCIENCE!

## THIS MONTH IN SCIENCE:

Would you want accelerated healing or surgery in a pill? Explore both, before moving on to understanding nature's weirdest reproductive practices as well as the possibility of the first self-assembling space telescope!



## Explore exoplanets

NASA has put up an interactive experience where users can get a feel of how the surface of alien worlds would appear. Users can take a virtual walk on the surface of Trappist-1 e and watch six Earth sized exoplanets more or less in a straight line in the sky. Kepler-186 f with its probably red vegetation, and Kepler-16 b with its twin suns can also be explored. Each planet has labelled features with scientific explanations.

<https://dgit.in/etb>

## WHAT'S NEW

# Measuring the human impact on the Earth's biomass

If you weighed all the chickens in farms around the world, the figure would be three times that of all the birds in the wild. Humans and livestock combined weight 20 times as much as all the other animals in the world put together. This is an indication of the kind of impact humans have had on the biomass of the planet. The combined weight of all mammals was six times more than what it is now, before humans. We are responsible for eradicating roughly half of the biomass on Earth. The main cause of this impact is the clearing of forest and the cutting of trees. 80 per cent of the total biomass of the planet is because of



vegetation. Although they are incredibly tiny, they are so numerous that 15 per cent of the biomass is due to bacteria. All the other organisms, including humans, constitute only 5 per cent of the biomass on Earth.

<https://dgit.in/biomass>

## Astronomers image black hole eating star

For the first time, astronomers have observed the moment when a supermassive black hole swallowed up a star that drifted too close, and ejected jets of extremely fast moving high energy particles. The spectacular stellar death is known as a tidal disruption event, and only a small number of these have been detected. This was the first time that such



an event was directly observed. The star was pulverised and formed a fast moving disc of material around the black hole. From the poles of the disc, there

were two jets where particles were moving at nearly the speed of light. The event was tracked through multiple astronomical instruments, including the Very Long Baseline Array (VLBA), the Lovell telescope and the Spitzer telescope. The event took place in a pair of colliding galaxies, together known as Arp 299, 15 million light years away.

<https://dgit.in/arp299>



## React to unlock devices

Researchers from the University of Buffalo have proposed a system that measures brainwaves and your reaction to certain pictures to unlock devices. This novel biometric security measure works best with two kinds of images - pictures of celebrities, or pictures of animals.

<https://dgit.in/ubrct>



## MIT develops tiny drone chip

MIT researchers have engineered a tiny chip for drones. The chip will be used to navigate microdrones, and is called "Navion". The chip can process information from inertial sensors, as well as images from a camera at the rate of 171 frames per second.

<https://dgit.in/mitdc>



## Oldest reptile fossil found

Palaentologists have identified a 240 million year old reptilian fossil as *Megachirella wachtleri*. It is the oldest known ancestor of snakes and lizards. The fossil pushes back the advent of the first snakes and lizards by 75 million years.

<https://dgit.in/lizfos>